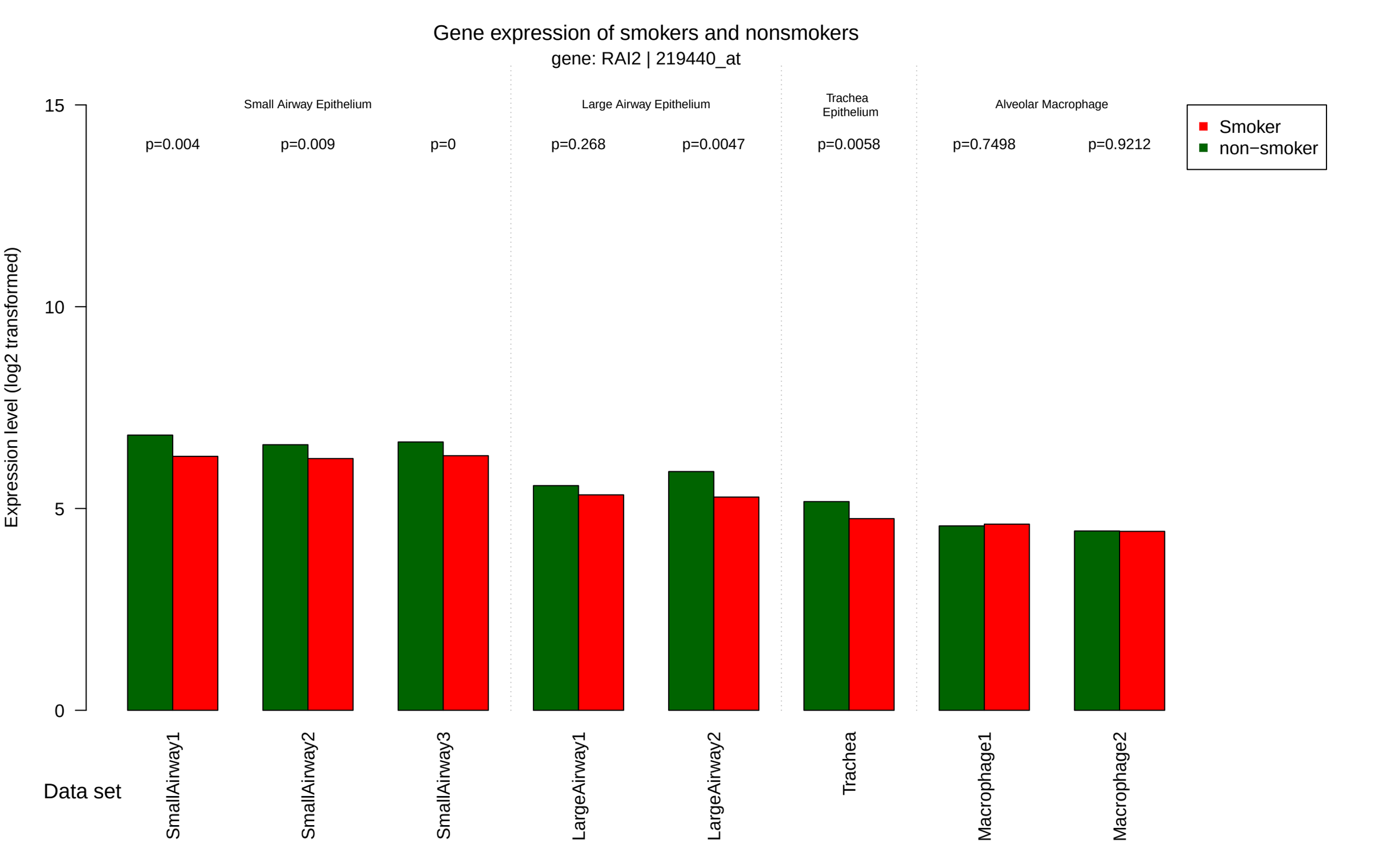
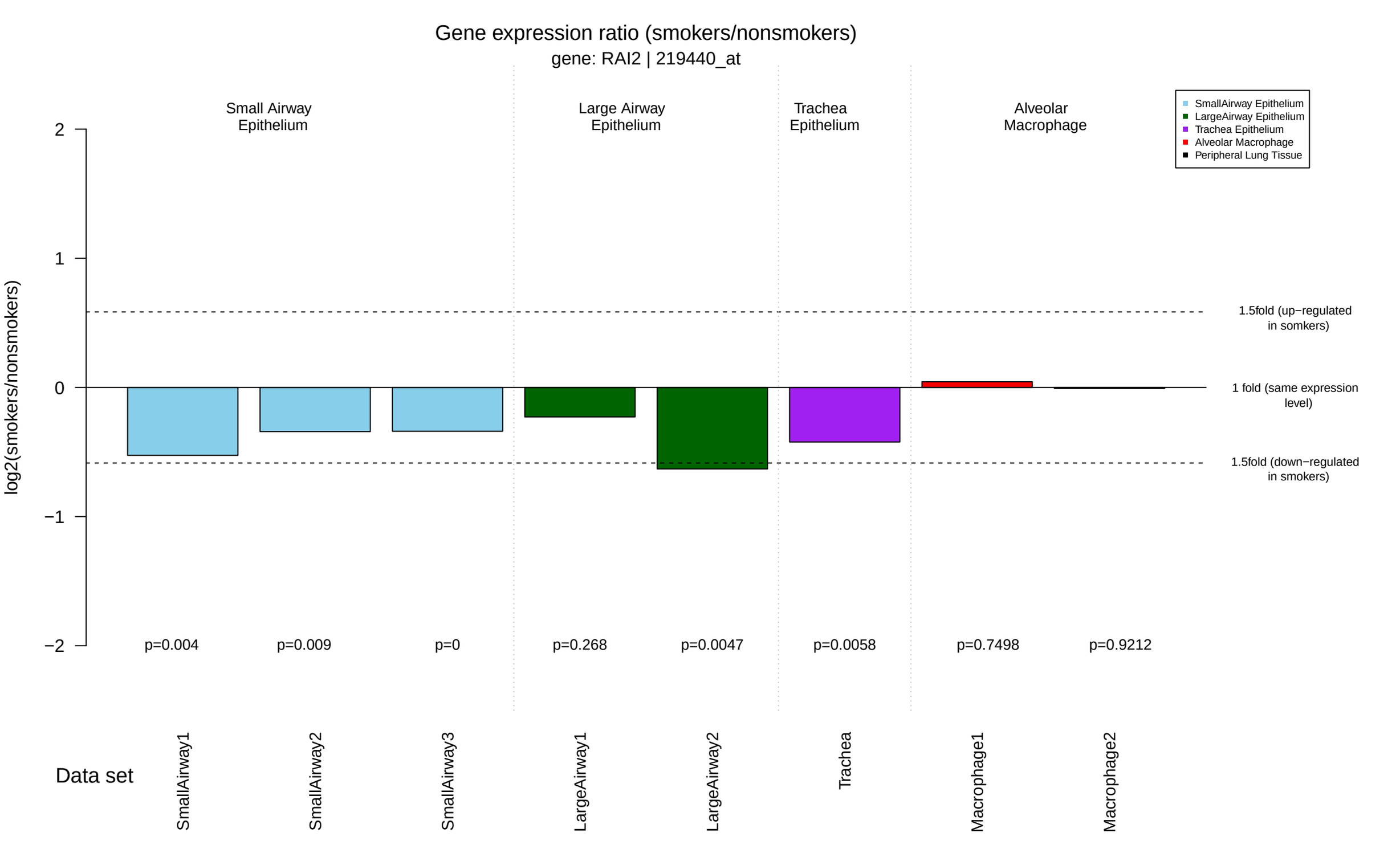
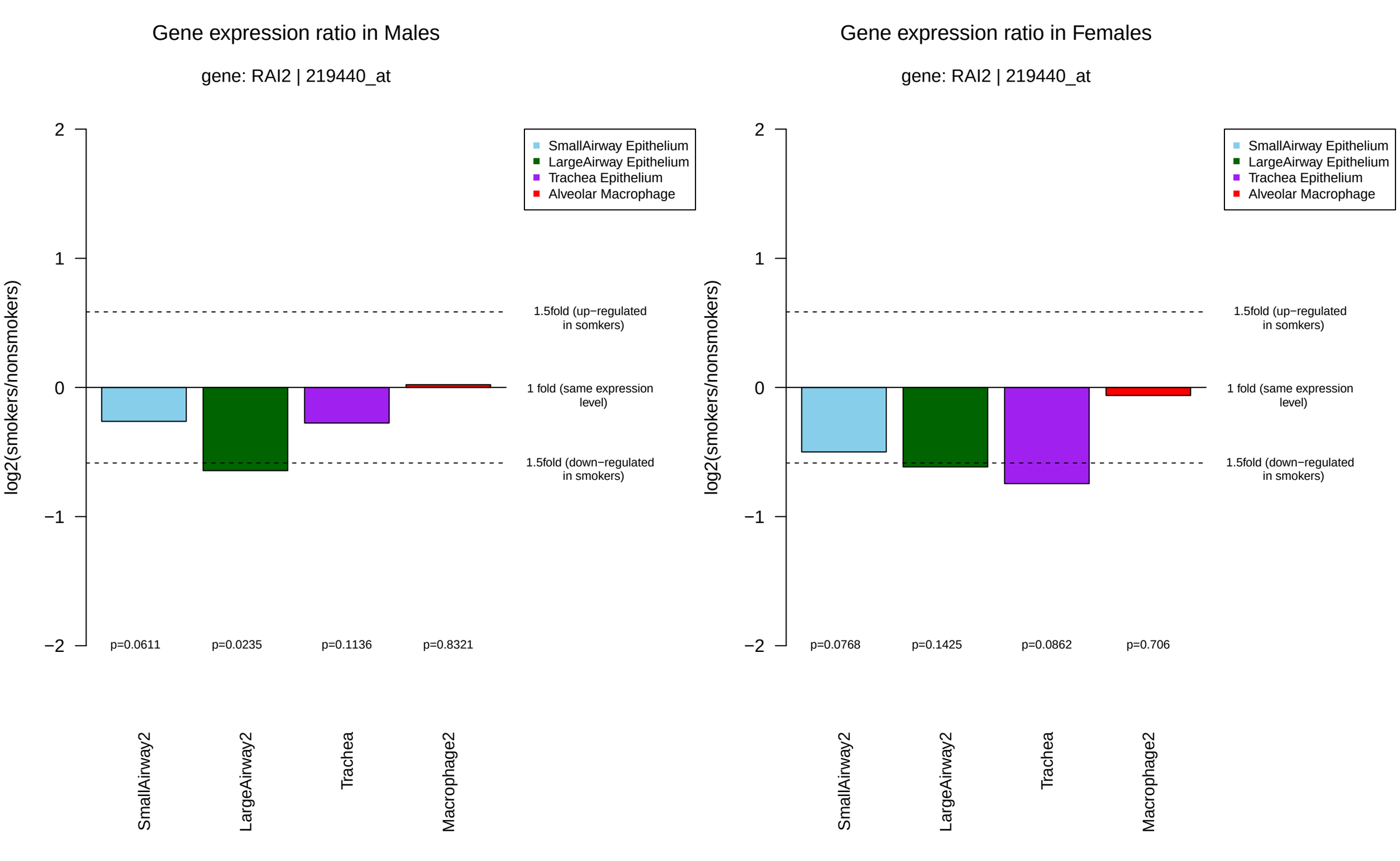
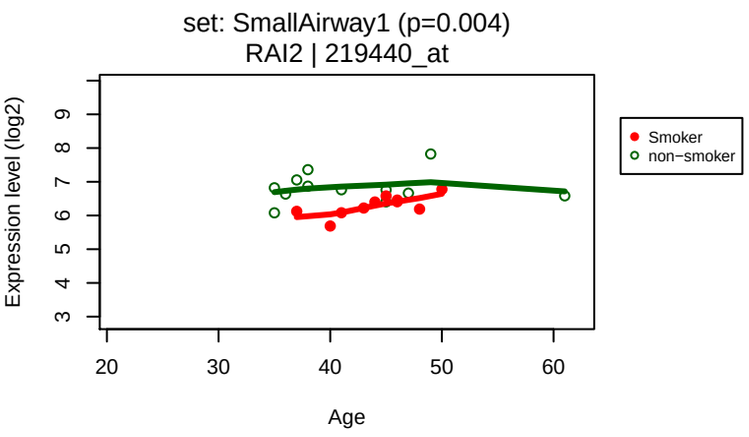
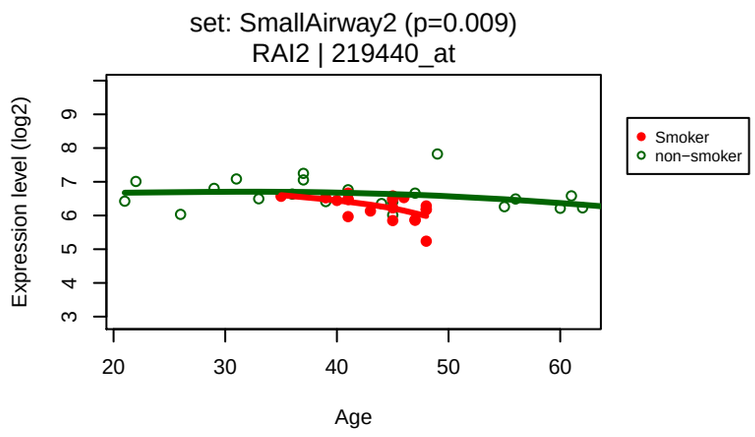
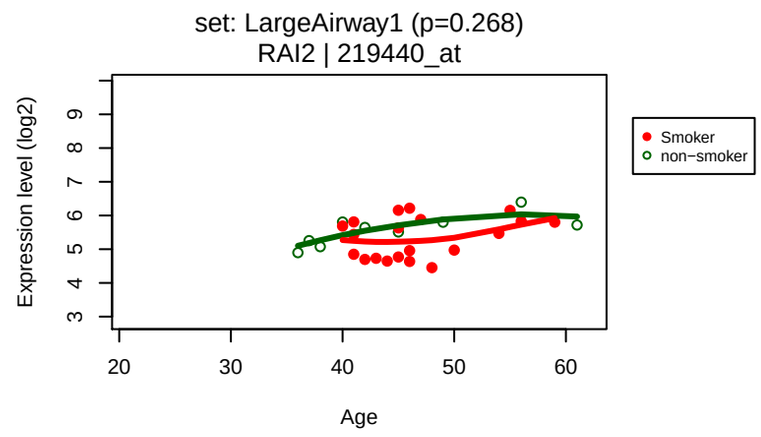

Gene expression of smokers and nonsmokers
gene: RAI2 | 219440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.004
p=0.009
p=0
p=0.268
p=0.0047
p=0.0058
p=0.7498
p=0.9212
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

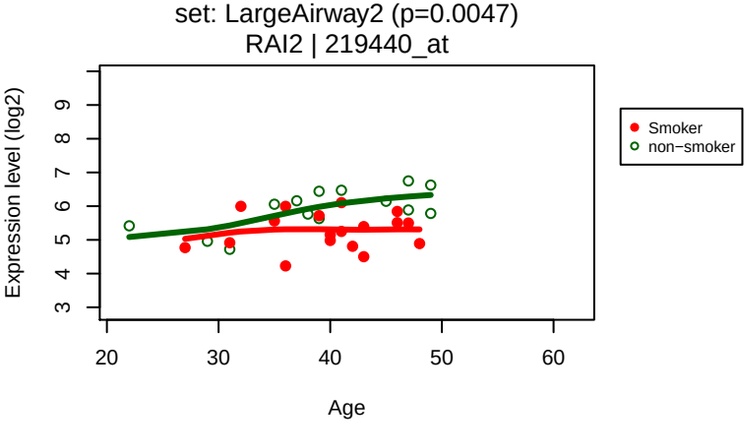

Gene expression ratio (smokers/nonsmokers)
gene: RAI2 | 219440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.004
p=0.009
p=0
p=0.268
p=0.0047
p=0.0058
p=0.7498
p=0.9212
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

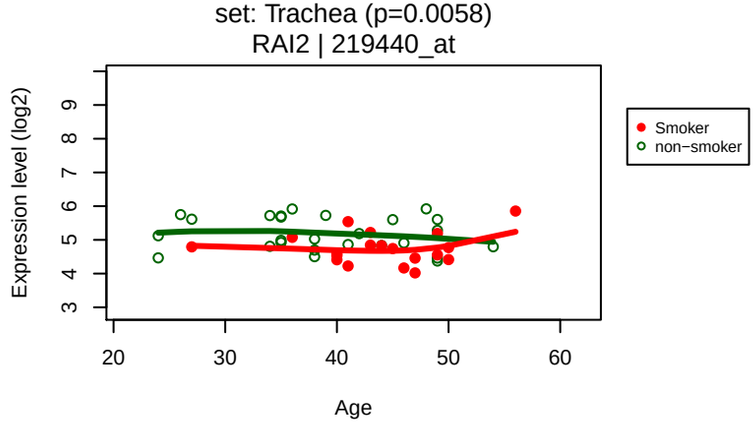

Gene expression ratio in Males
gene: RAI2 | 219440_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0611
p=0.0235
p=0.1136
p=0.8321
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAI2 | 219440_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0768
p=0.1425
p=0.0862
p=0.706
SmallAirway2
LargeAirway2
Trachea
Macrophage2

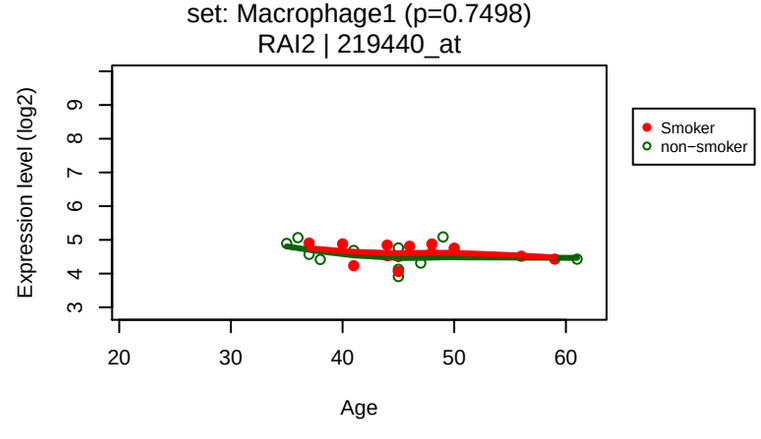

set: SmallAirway1 (p=0.004)
RAI2 | 219440_at
Expression level (log2)
Age
Smoker
non−smoker

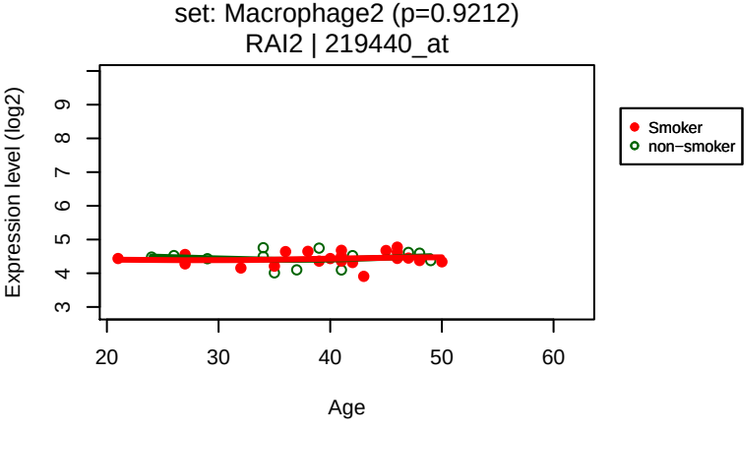

set: SmallAirway2 (p=0.009)
RAI2 | 219440_at
Expression level (log2)
Age
Smoker
non−smoker

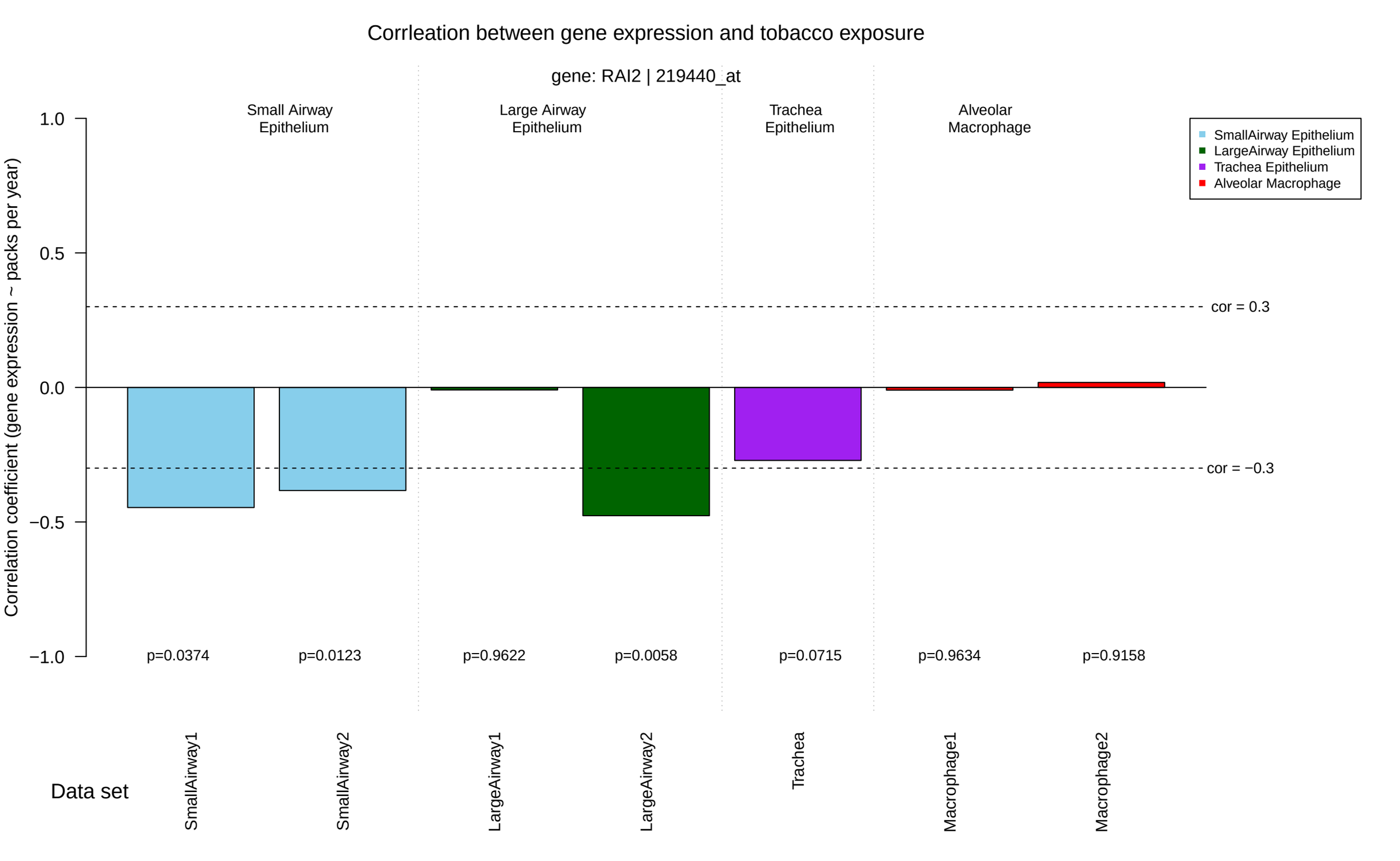

set: LargeAirway1 (p=0.268)
RAI2 | 219440_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0047)
RAI2 | 219440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0058)
RAI2 | 219440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7498)
RAI2 | 219440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9212)
RAI2 | 219440_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAI2 | 219440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0374
p=0.0123
p=0.9622
p=0.0058
p=0.0715
p=0.9634
p=0.9158
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2